

	Case Name : ACTIVIDADES GENERALES Project owner: Eder Omar Vilca Carrillo	Sector	Construction	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
	Risk Analysis	Project Control	Random simulation	
Processed by	Zahra Hussaini		One of nine std. scenarios	
Date	2025		User defined distributions	
File Name	C2025-16		Automatic tracking	
			Tracking based on user input	

1. Project description

Project authenticity

This project involves the construction and repair of multiple bridges—including Puente Conin, Puente Chapcar, and Puente Lanchan—through activities such as excavation, concrete work, gabion installation, and environmental mitigation, ensuring safe and sustainable infrastructure development. The project consists of activity, resource and cost data that were obtained directly from the actual project owner

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	49	Serial/Parallel (SP)	22%
Planned Duration (PD)	56 days	Activity Distribution (AD)	25%
Budget At Completion (BAC)	247278.93 EUR	Length of Arcs (LA)	0%
Renewable Resources	6 Types	Topological Float (TF)	39%
Consumable Resources	0 Types		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	13.7	12.3	166.7
CRI-rho	13.6	12.1	172.9
CRI-tau	9.4	8.6	146.5

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	26.0	8.6	360.2
CRI-rho	26.0	8.6	360.2
CRI-tau	26.0	8.6	360.2

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	24.5	42.5	122.3
SI	36.3	38.5	93.0
SSI	6.4	13.4	225.8
CRI-r	11.9	12.0	170.4
CRI-rho	11.6	12.1	164.9
CRI-tau	9.9	8.1	97.3

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	0.1	-0.1
PV - SPI	0.1	0
PV - SCI	0.1	0.1
ED - 1	0.1	-0.1
ED - SPI	0.1	0
ED - SCI	0.1	0
ES - 1	0.1	-0.1
ES - SPI(t)	0.1	0.1
ES - SCI(t)	0.1	0.1

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0.1	0.1
CPI	0.1	0.1
SPI	0.1	0.1
SPI(t)	0.1	0.1
SCI	0.1	0.1
SCI(t)	0.1	0.1
0.8 CPI + 0.2 SPI	0.1	0.1
0.8 CPI + 0.2 SPI(t)	0.1	0.1

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 1 tracking period. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	9586.8	0	-8	1	1	1	1
std dev	N/A	N/A	N/A	N/A	N/A	N/A	N/A
final	9586.8	0	-8	1	1	1	1

2.3.3.2. Time forecasting

PD	56
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Real Duration	57
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Late	1%
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EAC(t)	Real Accuracy			
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	55	N/A	0	0
PV - SPI	56	N/A	0	0
PV - SCI	57	N/A	0	0
ED - 1	58	N/A	0	0
ED - SPI	59	N/A	0	0
ED - SCI	60	N/A	0	0
ES - 1	61	N/A	0	0
ES - SPI(t)	62	N/A	0	0
ES - SCI(t)	63	N/A	0	0

2.3.3.3. Cost forecasting

BAC	247278.93 €
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Real Cost	247278.93 €
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On budget	0%
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EAC	Real Accuracy			
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	237692.1	N/A	0	0
CPI	237692.1	N/A	0	0
SPI	237692.1	N/A	0	0
SPI(t)	237692.1	N/A	0	0
SCI	237692.1	N/A	0	0
SCI(t)	237692.1	N/A	0	0
0.8 CPI + 0.2 SPI	237692.1	N/A	0	0
0.8 CPI + 0.2 SPI(t)	237692.1	N/A	0	0